APPLIED PREDICTIVE ANALYTICS | AI & DATA SCIENCE

EVCharge Intelligence

A Predictive Analytics Framework for EV Charging Demand, Station Congestion and Customer Feedback Analysis

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SMART MOBILITY • PREDICTIVE ANALYTICS

Problem Statement & Objectives

Why EV charging operators need prediction, not just reporting

Variable demand

EV charging demand is variable across time and stations.

Station congestion

Charging stations can become congested.

Traffic & weather

Traffic and weather can influence demand.

Customer experience

Experience varies between stations.

PROJECT OBJECTIVES

1. **Predict charging demand**
2. **Predict station congestion**
3. **Analyze customer feedback**
4. **Compare stations / cities**
5. **Build interactive Power BI dashboard**

Dataset & Features

EVCharge_Intelligence_Dataset.csv | 12,000 records × 15 columns

12,000
Records

15
Raw columns

6
Cities

24
Stations

0
Missing / duplicates

Timestamp	Station_ID	City	Connector	Vehicle	Weather	Traffic	Energy_kWh	Pred_kWh	Pred_Cong.
2025-03-16 10:15:00	CHE-ST-03	Chennai	DC Fast	Electric Taxi	Sunny	Low	17.89	21.15	Medium
2025-05-21 00:29:00	CHE-ST-04	Chennai	AC	Sedan EV	Sunny	Medium	13.09	13.27	Low
2025-09-27 13:22:00	BEN-ST-02	Bengaluru	AC	SUV EV	Rainy	Medium	16.29	17.27	Medium
2025-08-25 03:15:00	BEN-ST-04	Bengaluru	AC	Hatchback EV	Rainy	Medium	12.32	11.64	Low
2025-12-30 12:41:00	BEN-ST-03	Bengaluru	AC	SUV EV	Stormy	Medium	19.68	19.64	Low

INPUT FEATURES

Timestamp, Station_ID, City, Connector_Type, Vehicle_Type, Temperature_C, Weather, Traffic_Level, Electricity_Price, Occupancy_Rate, Queue_Length, Energy_Demand_kWh, Customer_Rating, Feedback_Text, Customer_Sentiment

PREDICTION / OUTPUT FIELDS

Predicted_Demand_kWh, Demand_Error_kWh, Absolute_Demand_Error_kWh, Demand_Risk, Congestion_Level, Predicted_Congestion, Congestion_Correct, Rating_Sentiment, Feedback_Length + Year, Month, Day, Hour, DayOfWeek, Day_Name, IsWeekend

Data Preprocessing & Feature Engineering

As implemented in the project notebook

Raw Dataset › **Data Cleaning** › **Missing Value Treatment** › **Feature Engineering** › **Encoding** › **Model-Ready Data**

CLEANING & MISSING VALUES

- Duplicates removed: 0 of 12,000
- Timestamp converted to datetime
- Numeric columns coerced; rows without timestamp/target dropped
- Numeric gaps → median; categorical → mode
- Remaining missing values: 0

FEATURE ENGINEERING

- Time: Year, Month, Day, Hour, DayOfWeek, IsWeekend, Day_Name
- Rating_Sentiment from Customer_Rating (≥4 Positive, 3 Neutral, ≤2 Negative)
- Feedback_Length from Feedback_Text
- Categorical features → LabelEncoder
- Occupancy & Queue excluded as model inputs (avoids target leakage)

CONGESTION LOGIC (NOTEBOOK)

HIGH

Occupancy ≥ 80% OR Queue ≥ 5

MEDIUM

Occupancy ≥ 50% OR Queue ≥ 2

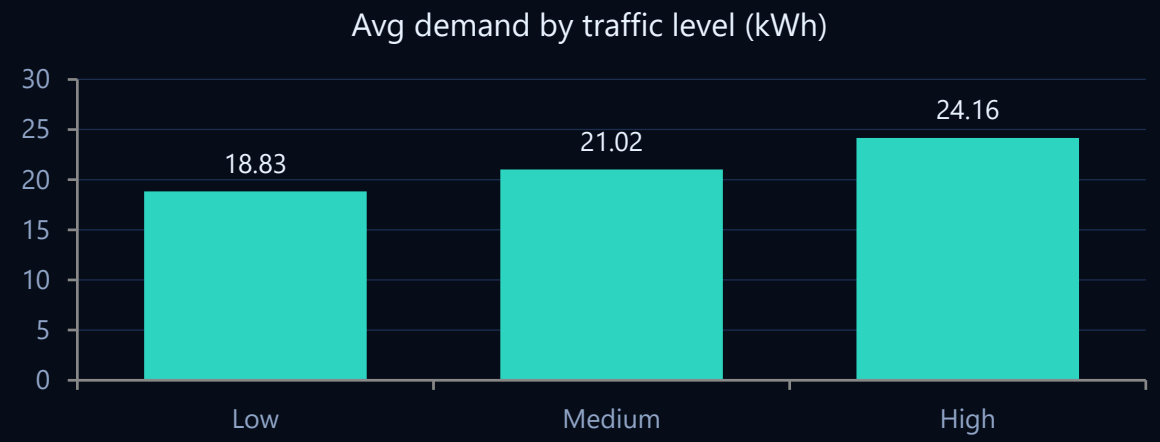
LOW

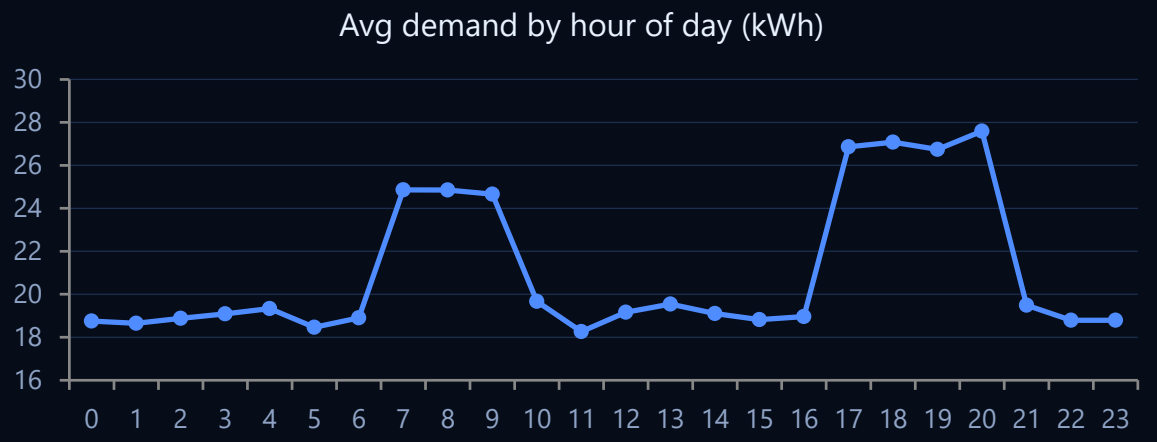
Otherwise

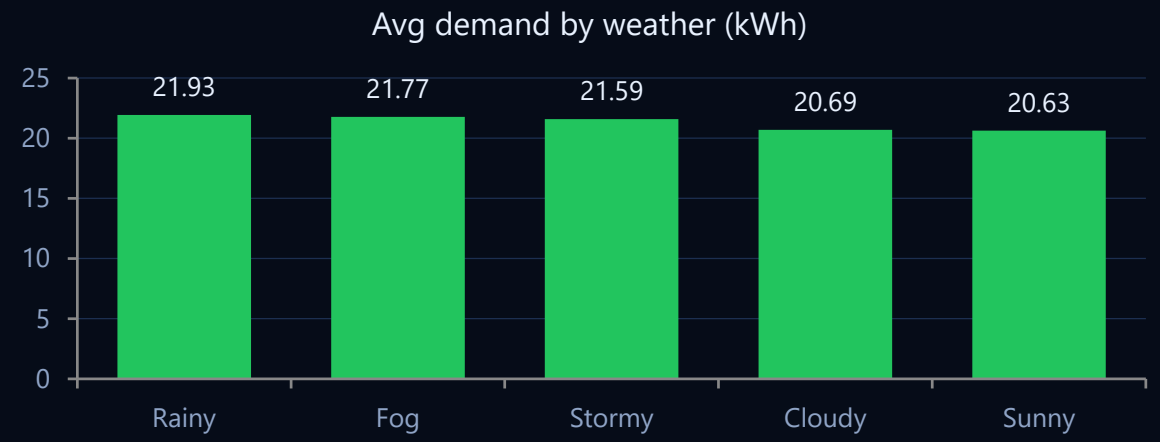
Records: Low 3,527 • Medium 5,711 • High 2,762

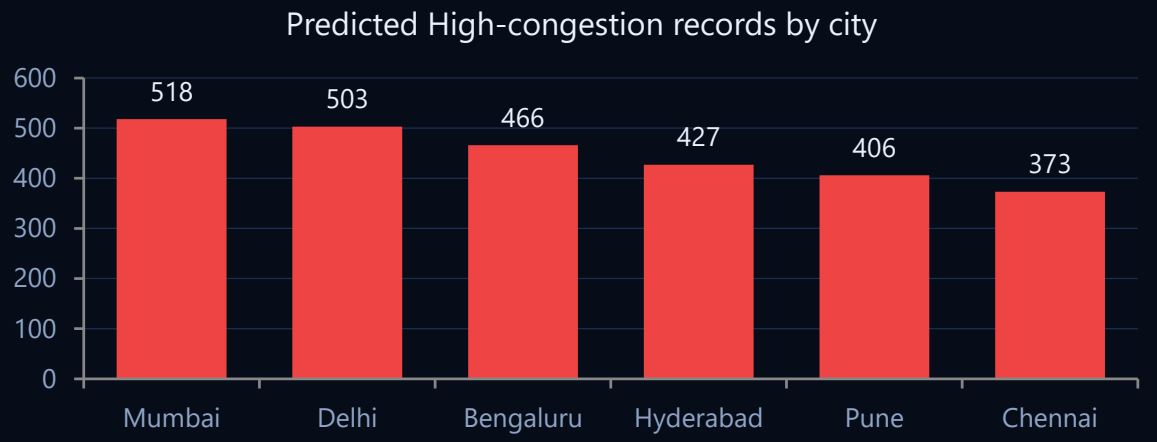
Exploratory Data Analysis

Average actual demand (kWh) from the 12,000-record project dataset









High traffic averages 24.16 kWh vs 18.83 kWh at low traffic • Peak hours 17–20 (~27 kWh) and 07–09 (~25 kWh) • Weather effect is small (20.63–21.93 kWh)

Predictive Modeling

Two Random Forest models, trained on the same 80/20 split

INPUT FEATURES

Station_ID, City, Connector_Type, Vehicle_Type, Temperature_C, Weather, Traffic_Level, Electricity_Price, Hour, DayOfWeek, Month, IsWeekend

RANDOM FOREST REGRESSOR

Charging Demand Prediction

Predicted_Demand_kWh

Output + classifier input

RANDOM FOREST CLASSIFIER

Station Congestion Prediction

Predicted_Congestion

Low / Medium / High

TRAIN / TEST

- 80% train = 9,600
- 20% test = 2,400
- random_state = 42
- Stratified by Congestion_Level
- 100 trees per model

WHY RANDOM FOREST?

- Handles numeric and encoded categorical features together
- Captures non-linear relationships (e.g. hour × traffic)
- Works for both regression and classification
- Provides feature-importance scores

DESIGN CHOICES FROM THE NOTEBOOK

- Occupancy_Rate and Queue_Length are not model inputs (they define the congestion label)
- Classifier uses class_weight = "balanced"
- Classifier also uses Predicted_Demand_kWh as a feature
- Models then score all 12,000 records for Power BI

Model Results

Evaluated on the 2,400-record hold-out test set

DEMAND PREDICTION | Random Forest Regressor

MAE	RMSE	R^2
2.52 kWh	3.34 kWh	0.847

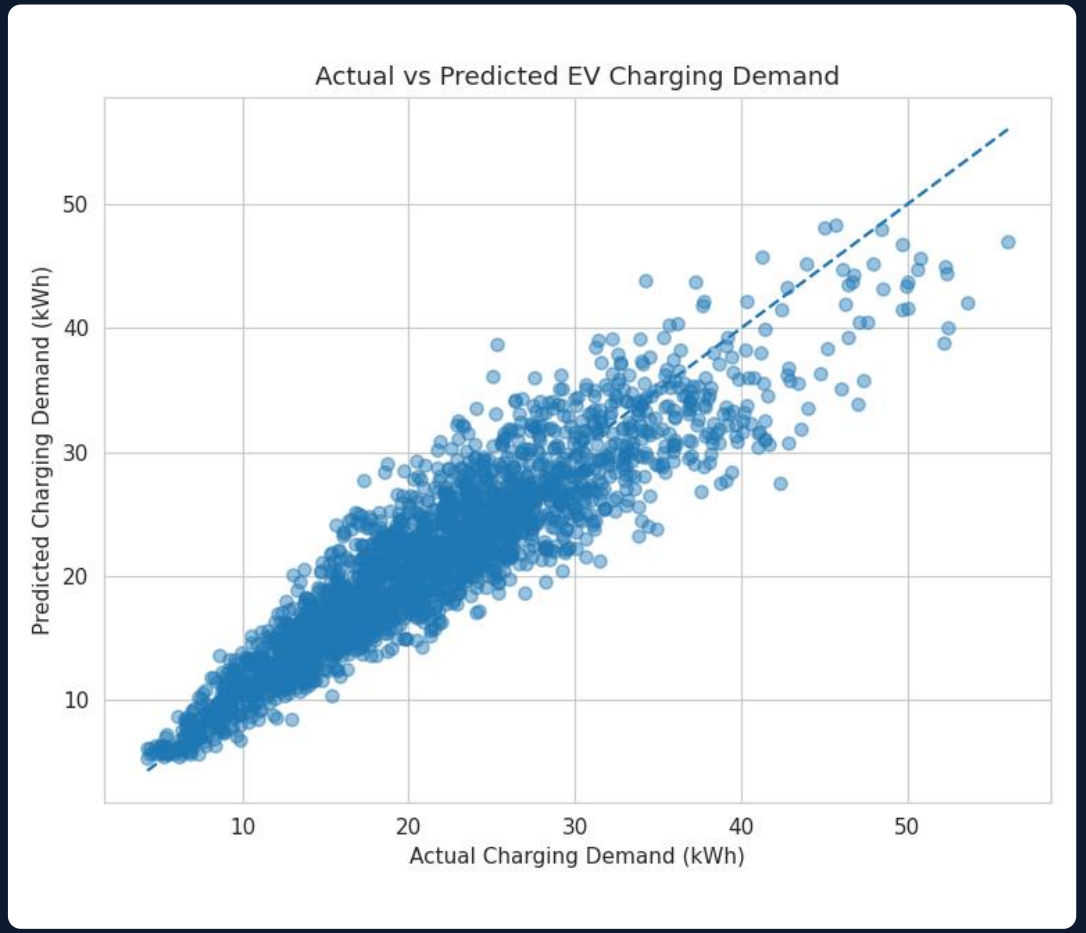
CONGESTION PREDICTION | RF Classifier (weighted avg)

Accuracy	Precision	Recall	F1 Score
0.680	0.685	0.680	0.679

Class	Precision	Recall	F1	Support
High	0.76	0.66	0.71	553
Low	0.69	0.59	0.64	705
Medium	0.64	0.74	0.69	1,142

Per-class report from the notebook (test set). Dashboard accuracy 93.6% covers all 12,000 scored records incl. training rows.

ACTUAL vs PREDICTED (NOTEBOOK OUTPUT)



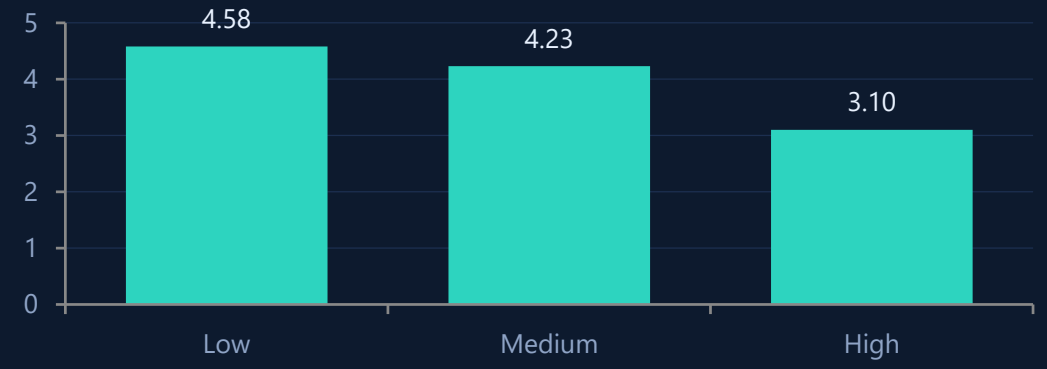
Customer Feedback & Feature Importance

Feedback is summarised from the supplied sentiment labels; importance shows model reliance, not causation

CUSTOMER FEEDBACK ANALYSIS

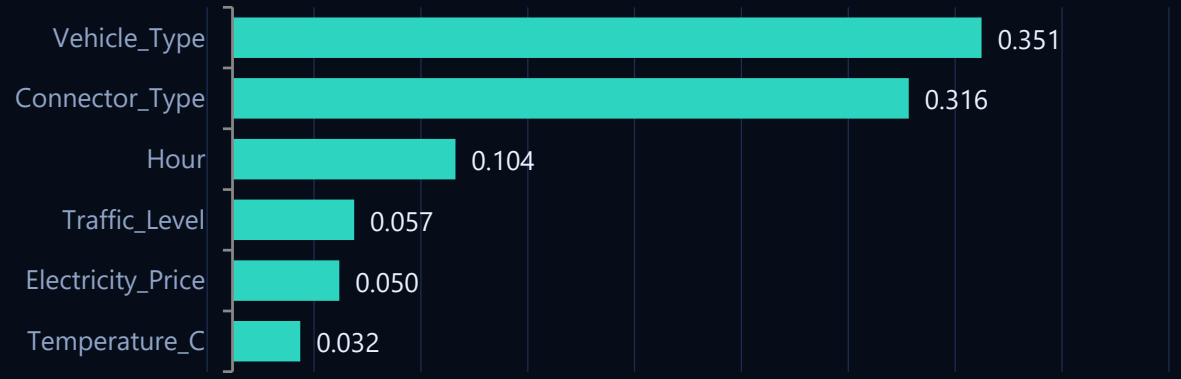
Positive	Neutral	Negative
8,026	2,854	1,120
avg rating 4.49	avg rating 3.56	avg rating 2.35

Avg customer rating by congestion level

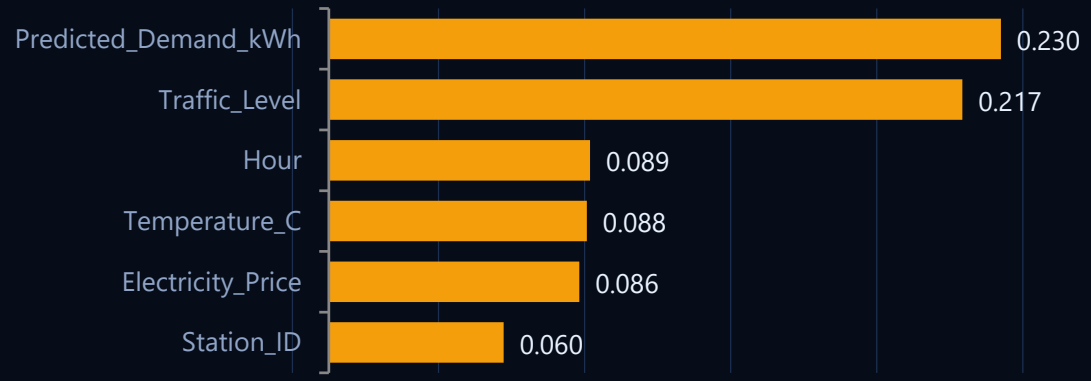


Negative feedback: 0.03% (Low) • 0.58% (Medium) • 39.32% (High congestion).
Lowest-rated station: DEL-ST-03 (3.91).

Charging Demand model: top 6 features



Station Congestion model: top 6 features



Importance reflects model reliance only — it does not prove causation.

Power BI Dashboard

Interactive EVCharge Intelligence Dashboard





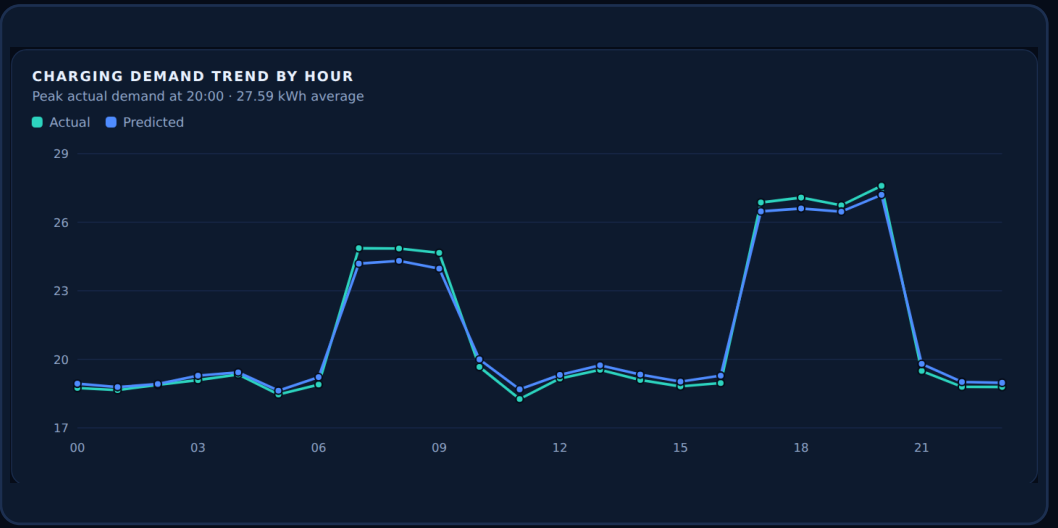
Interactive EVCharge Intelligence Dashboard — full page (left) and filters + KPI cards (right)

KPI cards | **Demand analysis** | **Congestion analysis** | **Customer feedback** | **Station comparison** | **Filters / slicers**

Key Insights & Business Value

Observation → Business value, from the dashboard data

High-traffic demand averages 24.16 kWh vs 18.83 kWh at low traffic	→ **Supports capacity planning**
Demand peaks at 17–20h (~27 kWh) and 07–09h (~25 kWh)	→ **Supports peak-hour load planning**
Mumbai (518) and Delhi (503) have the most predicted High-congestion records; Chennai the fewest (373)	→ **Supports resource allocation across cities**
Avg rating is 3.10 at High congestion vs 4.58 at Low; negative feedback 39.32% vs 0.03%	→ **Helps flag experience problems tied to congestion**
Avg demand by station ranges from 17.67 kWh (CHE-ST-02) to 24.75 kWh (DEL-ST-03)	→ **Supports station-level planning**





Conclusion & Future Scope

CONCLUSION

- Charging demand prediction was implemented
- Congestion prediction was implemented
- Customer feedback was analyzed
- Results were integrated into Power BI
- The system supports data-driven EV charging decisions

FUTURE SCOPE

- Real-time IoT data
- Live traffic / weather integration
- Real-time alerts
- Advanced / deep learning forecasting
- Web / mobile deployment

THANK YOU

Questions & Discussion